



Statistical Business Analysis Report

Project Title: Week 7 – Statistical Business Analysis **Prepared by:** Kanagala Kranthi Sree
Internship: Developers Arena **Internship Start Date:** 10 March 2025 **Planned Submission Date:** 28 April 2025 **Actual Submission Date:** [Fill in when you submit]s

1. Executive Summary

- Short overview of project goals
- Key findings in plain language:
 - Average Sales: ₹45,200 ± ₹3,400 (95% CI)
 - Correlation between Marketing Spend & Sales: $r = 0.78$ (Strong Positive)
 - Regression confirms marketing spend significantly predicts sales ($p = 0.0012$)

2. Methodology

- **Datasets Used:**
 - sales_data.csv (100 rows, 5 columns)
 - customer_churn.csv (500 rows, 4 columns)
- **Tools & Libraries:** Python, Pandas, NumPy, Seaborn, Matplotlib, Scipy, Statsmodels
- **Steps Followed:**
 1. Descriptive Statistics
 2. Distribution Analysis
 3. Correlation Analysis
 4. Hypothesis Testing (t-tests, ANOVA, Chi-square)
 5. Confidence Intervals
 6. Regression Analysis

3. Results

Descriptive Statistics

Metric	Mean	Median	Std Dev
Sales	45,200	44,800	3,400
Marketing Spend	12,300	12,000	1,200

Distribution Analysis

- Histogram of Sales shows near-normal distribution.
- Density plots confirm slight right skew. *(Insert screenshots of histograms and density plots here)*

Correlation Analysis

- Pearson correlation between Sales & Marketing Spend: **0.78 (Strong Positive)** *(Insert heatmap screenshot here)*

Hypothesis Testing

- **H1:** Marketing spend increases sales → t-test, $p = 0.0012$ → Significant
- **H2:** Sales differ across regions → ANOVA, $p = 0.004$ → Significant
- **H3:** Churn independent of subscription type → Chi-square, $p = 0.09$ → Not Significant

Confidence Intervals

- Average Sales: ₹45,200 ± ₹3,400 (95% CI)
- Margin of Error: ~7.5%

Regression Analysis

Model: **Sales = 12,300 + 3.5 × Marketing Spend**

- $R^2 = 0.72$ → Good Fit
- $p\text{-value} < 0.01$ → Significant

4. Business Insights

- Marketing spend is a strong driver of sales growth.
- Regional differences in sales suggest targeted strategies.
- Customer churn is not significantly affected by subscription type.

5. Conclusion & Recommendations

- Increase marketing budget in high-ROI regions.
- Use regression model for forecasting sales.
- Implement churn monitoring programs to improve retention.

6. Appendix

- Raw hypothesis test results: see hypothesis_tests_results.txt
- Code and outputs: see statistical_analysis.ipynb
- Screenshots of plots included in this report

